3.3.10

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Question

Construct a right triangle ABC with $AB = 6\,\text{cm}$, $BC = 8\,\text{cm}$ and $\angle B = 90^\circ$. Draw BD, the perpendicular from B on AC. Draw the circle through B, C and D and construct the tangents from A to this circle.

Theoretical Solution

Let us place the triangle in coordinate geometry using vectors. Let:

$$\mathbf{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 8 \end{pmatrix} \tag{1}$$

Since $\angle B = 90^\circ$, triangle ABC is right-angled at B.

Let $\mathbf{n} = \mathbf{a} - \mathbf{c} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 8 \end{pmatrix} = \begin{pmatrix} 6 \\ -8 \end{pmatrix}$

Theoretical Solution

Let the general point on line AC be:

$$\mathbf{r} = \mathbf{c} + \lambda(\mathbf{a} - \mathbf{c}) = \mathbf{c} + \lambda\mathbf{n} \tag{2}$$

The foot of the perpendicular from $\mathbf{b}$ to line $\mathbf{r}$ is the point on AC closest to $\mathbf{b}$.
Minimize the square of the distance:

$$\|\mathbf{r} - \mathbf{b}\|^2 = (\mathbf{c} + \lambda\mathbf{n} - \mathbf{b})^\top(\mathbf{c} + \lambda\mathbf{n} - \mathbf{b}) \tag{3}$$

Differentiate w.r.t. λ and set derivative to zero:

$$\frac{d}{d\lambda}\left[\|\mathbf{r} - \mathbf{b}\|^2\right] = 2\mathbf{n}^\top(\mathbf{c} + \lambda\mathbf{n} - \mathbf{b}) = 0 \tag{4}$$

Solving:

$$\mathbf{n}^\top(\mathbf{c} - \mathbf{b}) + \lambda\mathbf{n}^\top\mathbf{n} = 0 \tag{5}$$

Compute terms:

Theoretical Solution

$$\mathbf{n}^\top(\mathbf{c}-\mathbf{b}) = \begin{pmatrix} 6 & -8 \end{pmatrix} \begin{pmatrix} 0 \\ 8 \end{pmatrix} = -64 \tag{6}$$

$$\mathbf{n}^\top \mathbf{n} = \begin{pmatrix} 6 & -8 \end{pmatrix} \begin{pmatrix} 6 \\ -8 \end{pmatrix} = 36 + 64 = 100 \tag{7}$$

Substitute into 6:

$$-64 + 100\lambda = 0 \Rightarrow \lambda = \frac{64}{100} = \frac{16}{25} \tag{8}$$

Therefore, point $\mathbf{d}$ is:

$$\mathbf{d} = \mathbf{c} + \frac{16}{25}(\mathbf{a}-\mathbf{c}) = \begin{pmatrix} 0 \\ 8 \end{pmatrix} + \frac{16}{25} \begin{pmatrix} 6 \\ -8 \end{pmatrix} = \begin{pmatrix} \frac{96}{25} \\ 8 - \frac{128}{25} \end{pmatrix} = \begin{pmatrix} \frac{96}{25} \\ \frac{72}{25} \end{pmatrix} \tag{9}$$

Let this be the circumcircle of triangle $\triangle BCD$.

Theoretical Solution

Use determinant form to find center $\mathbf{o}$ of the circle:

Circumcenter of $\triangle BCD$ lies at the intersection of perpendicular bisectors.

Let $\mathbf{o}$ be the center of the circle and r its radius. Then the tangents from $\mathbf{a}$ satisfy:

$$\|\mathbf{a} - \mathbf{o}\|^2 = r^2 + t^2 \tag{10}$$

Let $\mathbf{t}_1, \mathbf{t}_2$ be points of tangency. The tangent points lie on circle, and vectors $\mathbf{t}_i - \mathbf{o}$ are perpendicular to $\mathbf{a} - \mathbf{t}_i$, so again orthogonality condition applies:

$$(\mathbf{t}_i - \mathbf{o})^\top (\mathbf{a} - \mathbf{t}_i) = 0 \tag{11}$$

This gives quadratic system in coordinates of $\mathbf{t}_i$, which can be solved numerically or geometrically as per need.

Theoretical Solution

1. Plot $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{a} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 0 \\ 8 \end{pmatrix}$
2. Draw triangle ABC, right-angled at B
3. Compute $\mathbf{d}$ as in 10
4. Draw circle through $\mathbf{b}, \mathbf{c}, \mathbf{d}$
5. Draw tangents from point $\mathbf{a}$ to the circle using construction

Plot

beamer/figs/matg6.jpeg